Data Management

Table of contents

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.1     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```r
library(descr)
library(knitr)
library(dplyr)
library(readr)
library(readxl)
library(ggplot2)
```

===My Data===

```r
file_path <- "/cloud/project/data/data.xls"

data.xls <- read_excel(file_path)
options(width=80)
head(data.xls)
```

```
# A tibble: 6 x 138
  RespondentID Country  A3.1  A3.2  A3.3  A3.4  A3.5  A3.6  A3.7  A3.8  A3.9
         <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1            1       1     2     3    52    52     4     1     5    12     1
```

```
2            2       1     2     5     52    52     3     1     8     2     5
3            3       1     1     4     94    94     4     1     5     2     1
4            4       1     1     2    221   221     1     1    10     3     4
5            5       1     1     3     52    52     3     1     8     2     0
6            6       1     1     1     52    52     1     4     9     3     4
# i 127 more variables: A3.10 <dbl>, A5.1 <dbl>, A5.2 <dbl>, `A5.3*` <dbl>,
#   A5.4 <dbl>, A5.5 <dbl>, A6.1 <dbl>, A7.1 <dbl>, A7.2 <dbl>, A9.1 <dbl>,
#   A9.2 <dbl>, `A9.3*` <dbl>, A9.4 <dbl>, A9.5 <dbl>, A9.6 <dbl>, A9.7 <dbl>,
#   `A9.8*` <dbl>, A11.1 <dbl>, A11.2 <dbl>, A11.3 <dbl>, A11.4 <dbl>,
#   A12.1 <dbl>, A12.2 <dbl>, A12.3 <dbl>, A12.4 <dbl>, A13.1 <dbl>,
#   A13.2 <dbl>, A13.3 <dbl>, A13.4 <dbl>, A14.1 <dbl>, A14.2 <dbl>,
#   A14.3 <dbl>, A14.4 <dbl>, A16.1 <dbl>, A16.2 <dbl>, `A16.3*` <dbl>, ...
```

===Frequencys===

```
freq(as.ordered(data.xls$A3.8), plot = FALSE)
```

```
as.ordered(data.xls$A3.8)
      Frequency  Percent Cum Percent
1           293  14.4264       14.43
2           566  27.8680       42.29
3            87   4.2836       46.58
4            22   1.0832       47.66
5            11   0.5416       48.20
6            10   0.4924       48.70
7            11   0.5416       49.24
8            15   0.7386       49.98
9            55   2.7080       52.68
10          155   7.6317       60.32
11           75   3.6928       64.01
12          731  35.9921      100.00
Total      2031 100.0000
```

===Frequency of Religious Orientations===

```
freq(as.ordered(data.xls$A3.8), plot = FALSE)
```

```
as.ordered(data.xls$A3.8)
      Frequency  Percent Cum Percent
1           293  14.4264       14.43
2           566  27.8680       42.29
3            87   4.2836       46.58
4            22   1.0832       47.66
5            11   0.5416       48.20
6            10   0.4924       48.70
7            11   0.5416       49.24
```

```
8             15   0.7386       49.98
9             55   2.7080       52.68
10           155   7.6317       60.32
11            75   3.6928       64.01
12           731  35.9921      100.00
Total       2031 100.0000
```

===Frequency of Levels of Educations===

```
freq(as.ordered(data.xls$A3.10), plot = FALSE)
```

```
as.ordered(data.xls$A3.10)
      Frequency Percent Cum Percent
1            49   2.413       2.413
2           323  15.903      18.316
3           330  16.248      34.564
4           565  27.819      62.383
5           441  21.713      84.097
6           252  12.408      96.504
7            37   1.822      98.326
8            34   1.674     100.000
Total      2031 100.000
```

This frequency table shows the levels of education within a sample. It shows me how many people don't have a high school degree to how many people have a doctorates. This will help me relate the info to people who are religious,to their levels of education. I need this info because I will eventually compare the information I gather to how many took COVID-19 as a serious threat.

===Frequency of who believed COVID-19 was exaggerated===

```
freq(as.ordered(data.xls$`A5.3*`), plot = FALSE)
```

```
as.ordered(data.xls$`A5.3*`)
      Frequency Percent Cum Percent
1           592  29.148       29.15
2           401  19.744       48.89
3           259  12.752       61.64
4           168   8.272       69.92
5           148   7.287       77.20
6           139   6.844       84.05
7           122   6.007       90.05
8            94   4.628       94.68
9            60   2.954       97.64
10           48   2.363      100.00
Total      2031 100.000
```

This frequency table shows the levels in which how many people thought COVID-19 was being exaggerated. It goes from level 1 = Not at all to level 10= Extremely. This will help me see how seriously people took COVID-19 and who underestimated it.

===Frequency in which mask was worn===

```r
freq(as.ordered(data.xls$A31.4), plot = FALSE)
```

```
as.ordered(data.xls$A31.4)
      Frequency Percent Cum Percent
1           874  43.033       43.03
2           121   5.958       48.99
3           120   5.908       54.90
4           185   9.109       64.01
5           184   9.060       73.07
6           195   9.601       82.67
7           352  17.331      100.00
Total      2031 100.000
```

This frequency table shows how many people had frequently put on a mask during COVID-19. It goes from level 1, Never, to level 7, Always. This will help aid my question on who was taking COVID-19 seriously.

===Distance Away Frequency===

```r
freq(as.ordered(data.xls$A31.6), plot = FALSE)
```

```
as.ordered(data.xls$A31.6)
      Frequency  Percent Cum Percent
1            24   1.1817       1.182
2            20   0.9847       2.166
3            36   1.7725       3.939
4           106   5.2191       9.158
5           138   6.7947      15.953
6           496  24.4215      40.374
7          1211  59.6258     100.000
Total      2031 100.0000
```

This table shows the frequency in which a person kept a distance of 1.5 meters away from others. The chart goes from level 1, Never, to level 7, Always. This will help me see who didn't wear their masks during COVID-19, which relates to my question of who didn't follow the guidelines provided to everyone across the globe.

=== Variable for Levels of Education===

```r
data.xls$LVL_EDU <- factor(ifelse(data.xls$A3.10 %in% c("1","2"), "High School and Below", "Co

summary(data.xls$LVL_EDU)
```

```
   College and Above High School and Below
                1659                   372
```

I am creating a variable named LVL_EDU to help me see who have educations from up to High School then College and higher.

===Variable for People of Only Religious Orientations===

```r
data.xls$ONLY_R <- factor(ifelse(data.xls$A3.8 %in% c("1", "2","3","4","5","6","7","8","9","11

summary(data.xls$ONLY_R)
```

```
Has a Religious Belief             No Beliefs
                  1145                    886
```

I am creating a variable named ONLY_R to group up who and who don't have a religious orientation.

===Variable for People Who Think COVID-19 was over exaggerated===

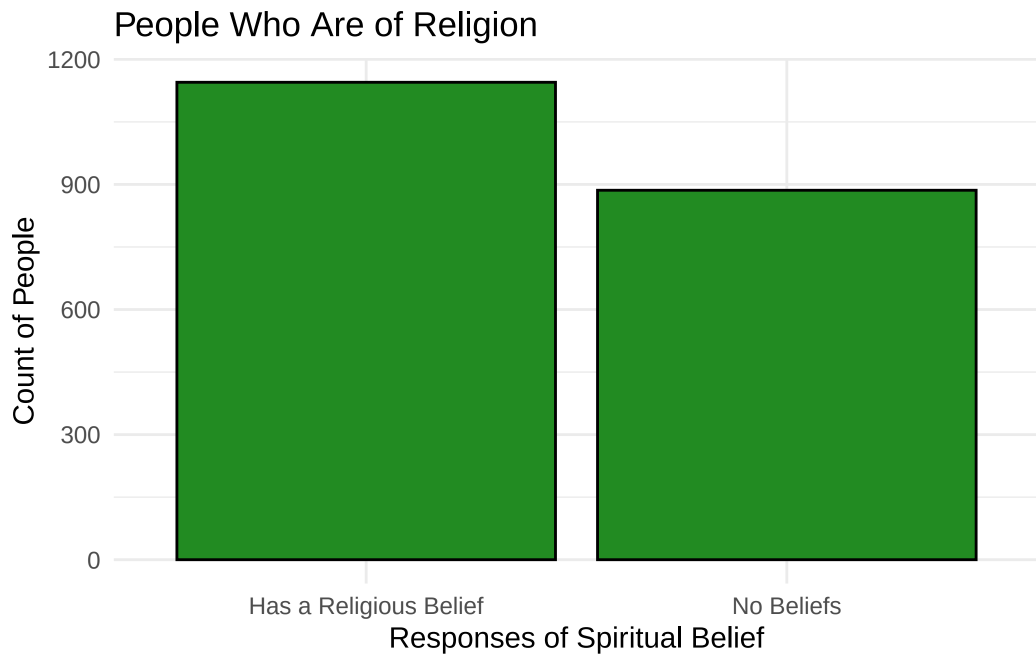
```r
data.xls$CVD_EXAG <- factor(ifelse(data.xls$`A5.3*` %in% c("1", "2","3","4","5"), "Reasonable C

summary(data.xls$CVD_EXAG)
```

```
 Out-of-Proportion Reasonable Concern
               463               1568
```

I am creating the variable CVD_EXAG to summarize the responses 1 = Not at all, 10= Extremely into two managed variables, options 1-5 being considered the people who believed that COVID-19 was not being over-exaggerated and options 6-10 being considered the people who thought the virus was being told to be a larger issue than it really was.

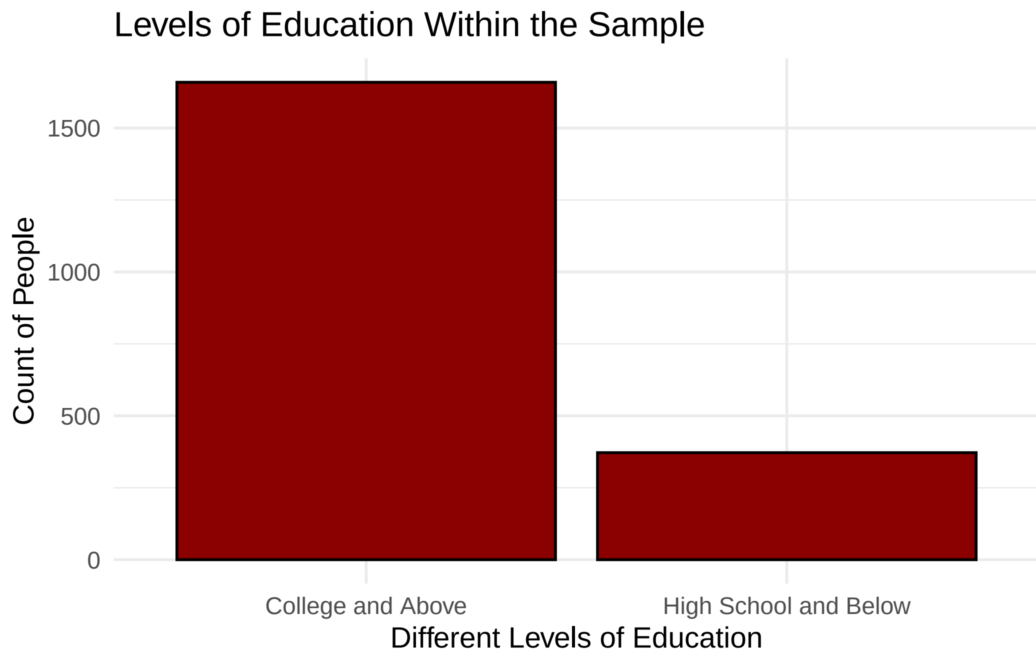
===Univariate Graphs===

```r
library(ggplot2)

data.xls %>%
  ggplot(aes(x = ONLY_R)) +
  geom_bar(fill = "forestgreen", color = "black") +
  ggtitle("People Who Are of Religion") +
  xlab("Responses of Spiritual Belief") +
  ylab("Count of People") +
  theme_minimal()
```



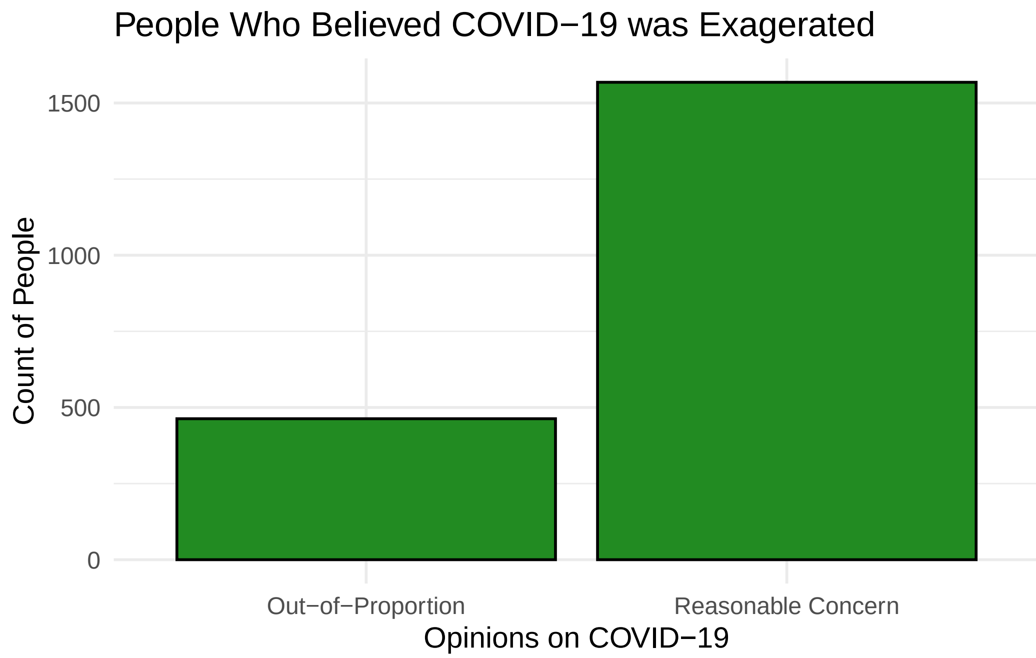
I am developing this graph to show how many people in the sample do and do not have any religious perspectives. I would like to correlate this graph later on with my other variables to see if there are any patterns between each other. One of my main concerns is that some people of religious backgrounds didn't think COVID-19 was a problem and that nothing would happen to each other. With this data gathered and graphed, I think it'll become easier to start comparing.

```
library(ggplot2)

data.xls %>%
  ggplot(aes(x = LVL_EDU)) +
  geom_bar(fill = "darkred", color = "black") +
  ggtitle("Levels of Education Within the Sample") +
  xlab("Different Levels of Education") +
  ylab("Count of People") +
  theme_minimal()
```



This graph will help to visualize the differences between two major levels of education. The goal of this is to understand whether or not the levels education dictate what people believed during the COVID-19 pandemic and how serious the virus was. My graph here is just to show how many in the sample have received higher education or not.

```
library(ggplot2)

data.xls %>%
  ggplot(aes(x = CVD_EXAG)) +
  geom_bar(fill = "forestgreen", color = "black") +
  ggtitle("People Who Believed COVID-19 was Exagerated") +
  xlab("Opinions on COVID-19") +
  ylab("Count of People") +
  theme_minimal()
```



This graph shows the many people who believed whether COVID-19 was overtly told as a massive issue and the many others who believed the concern of the virus was authentic. The main goal for this chart is to get an idea of the opinions many had about the virus when the issue was most prevalent.

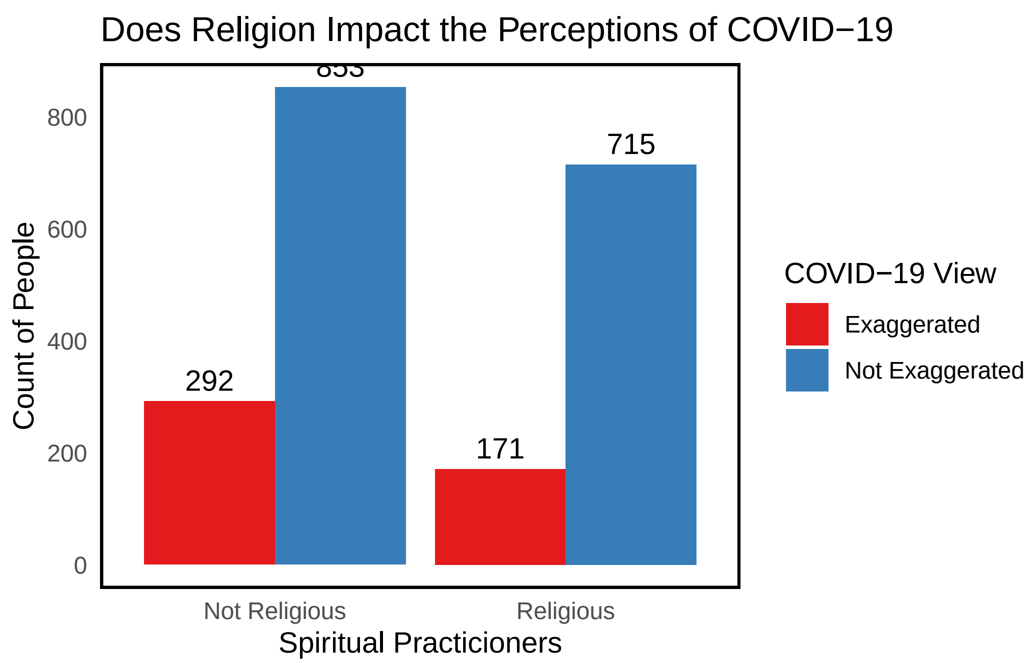
====Bivariate Graph====

===Is Religion a Cause of Disregarding the COVID-19 Virus?===

```
ggplot(data=data.xls, aes(x = factor(ONLY_R, labels = c("Not Religious", "Religious")), fill =
    geom_bar(position = "dodge") +
    geom_text(stat='count', aes(label=..count..), position=position_dodge(0.9), vjust=-0.5) +
    labs(title = "Does Religion Impact the Perceptions of COVID-19",
         x = "Spiritual Practicioners",
         y = "Count of People",
         fill = "COVID-19 View") +
    theme_minimal() +
    scale_fill_brewer(palette = "Set1") +
    theme(panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(),
      panel.border = element_rect(color = "black", fill = NA, size = 1))
```

```
Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
i Please use the `linewidth` argument instead.
```

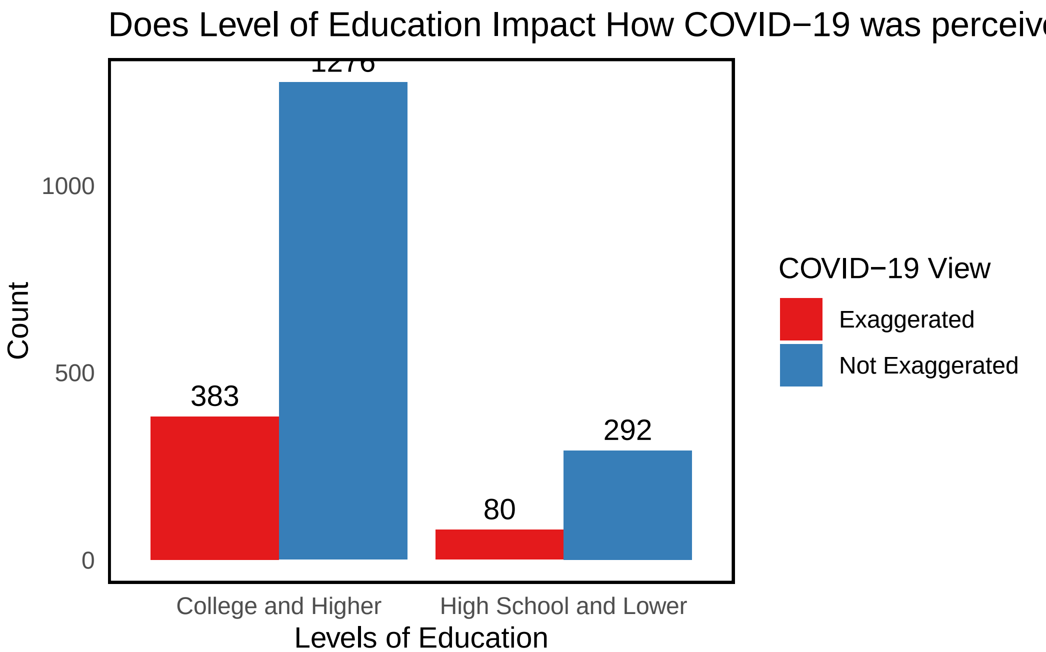
```
Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
i Please use `after_stat(count)` instead.
```



Through my findings I have found out that regardless of religion, a majority within a sample have agreed that COVID-19 was as serious as it was. I wanted to make this graph because through my own experiences during the pandemic, I witnessed many people disregarding the virus and explaining that their place of worship denied the virus being real or even an issue. This graph helps me see that it isn't really like this everywhere else.

===Did the Level of Education Affect the Perception of COVID-19 as a serious issue?===

```
ggplot(data=data.xls, aes(x = factor(LVL_EDU, labels = c("College and Higher", "High School and
    geom_bar(position = "dodge") +
    geom_text(stat='count', aes(label=..count..), position=position_dodge(0.9), vjust=-0.5) +
    labs(title = "Does Level of Education Impact How COVID-19 was perceived?",
         x = "Levels of Education",
         y = "Count",
         fill = "COVID-19 View") +
    theme_minimal() +
    scale_fill_brewer(palette = "Set1") +
    theme(panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(),
      panel.border = element_rect(color = "black", fill = NA, size = 1))
```



A common issue that's been witnessed among many is how a lack of education can lead people to be ignorant about certain theories, ideas or health issues. One of the most leading issues would be people who practice anti-vaxxing, the act of refusing to use a vaccine for medical problems. The cause of this is usually the lack of education to help a person understand why they should be taking these medical solutions. This is one of the biggest barriers in vaccine uptake. I wanted to see if the differences in education within the sample saw COVID-19 as an issue or not. Through my research and the data provided we can see that many, highly educated or not, did see the virus as an issue to take seriously.

===Chi-Square Test===

```r
data.xls <- read_excel(file_path)

data.xls$`A5.3*` <- as.factor(data.xls$`A5.3*`)
data.xls$A3.8 <- as.factor(data.xls$A3.8)

contingency_table <- table(data.xls$`A5.3*`, data.xls$A3.8)

chi_test <- chisq.test(contingency_table)
```

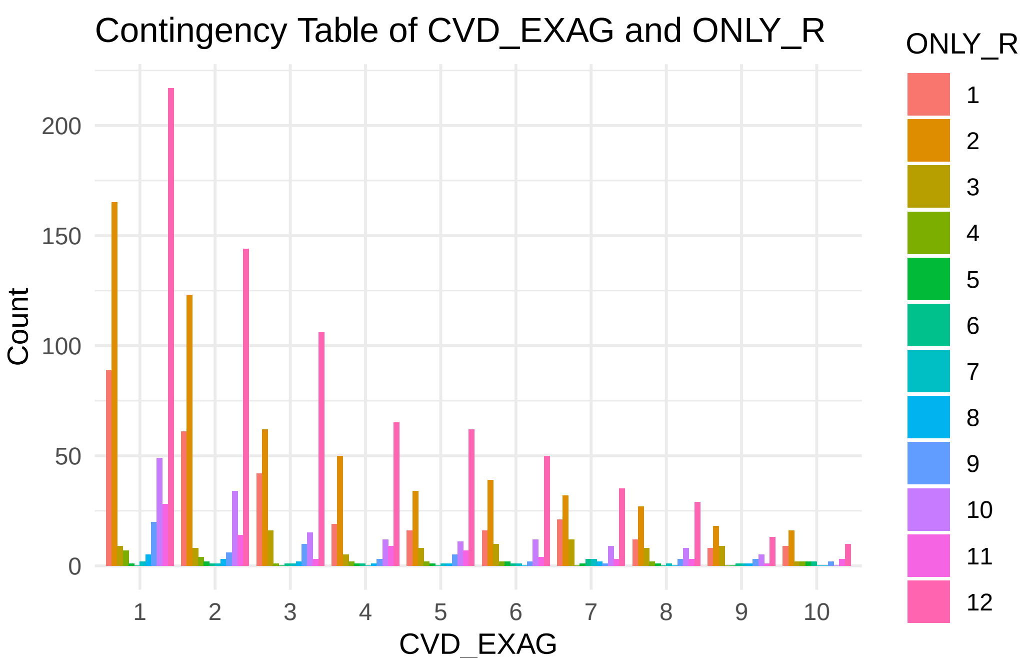
```
Warning in chisq.test(contingency_table): Chi-squared approximation may be
incorrect
```

```r
print(chi_test)
```

```

	Pearson's Chi-squared test
```

```
data:  contingency_table
X-squared = 171.32, df = 99, p-value = 8.897e-06
```

```r
plot_data <- as.data.frame(contingency_table)
colnames(plot_data) <- c("CVD_EXAG", "ONLY_R", "Count")

ggplot(plot_data, aes(x = CVD_EXAG, y = Count, fill = ONLY_R)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Contingency Table of CVD_EXAG and ONLY_R",
       x = "CVD_EXAG",
       y = "Count",
       fill = "ONLY_R") +
  theme_minimal()
```



The chi-square test I conducted resulted with 171.32 for X-Squared, 99 for df, and a p-value of 8.897e-06. This shows that there is a significant association between the respondents spirituality and their perception of COVID-19 being exaggerated. The p-value is much smaller than the conventional alpha level of 0.05, so we reject the null hypothesis of independence. This implies that a persons spiritual views may alter how they view COVID-19 as a serious issue, although more analysis would need to be conducted in order to understand this relationship more.